

Announcements

None

Announcements

- More readable code coming soon
- Github markdown via Quarto .qmd

Today

- Using modern RNG libraries
- Events in continuous time
- Exponential distribution

Modern algorithms

1. generate random bits
2. turn those into integers
3. divide by largest integer to give
 $\text{Uniform}(0,1)$

Modern algorithms

- **R:** Mersenne_Twister
- **Numpy:** PCG64 (Permuted Congruential Generator)

Discrete vs continuous time

- Discrete: all events in a period
- Continuous: each event occurs at a particular time
 - i.e. nanosecond

Events in continuous time

- When does the event occur?
 - “time to event”
 - “waiting time”

When does the individual die?

- Approximation
 - small interval of time, Δt
 - constant probability of death, p_{death}
- Diagrams
 - Event sequence diagram
 - Flowchart

When does the individual die?

Algorithm

set m , Δt , $p_{\text{death}} = m\Delta t$

set state = alive, $t = 0$

while state = alive

$t = t + \Delta t$

 if death occurs given Bernoulli(p_{death})

 state = dead

print t (time to death)

Bernoulli process